DEMO OPERATOR CO. (60% OPERATED JV)

DIRECTION CENTRALE HSE & DÉVELOPPEMENT DURABLE

BASIN ALPHA OIL & GAS COMPLEX (FACILITY A-1)

Operational Greenhouse Gas Inventory & Decarbonization Statement (2023 – 2025)

Reporting Period:	2023 – 2025 (Annual Multi-Year Audited Series)	**Accounting Standard:**	ISO 14064-1:2018 / GHG Protocol
Operating Unit:	Demo Operator Co. (60% Operated JV)	**GWP Metric Applied:**	IPCC AR5 (CH4=28.0)
Target Basin:	Basin Alpha (Block 101)	**Statutory Mandate:**	Executive Decree 21-330 & Decree 06-138
Monitored Facilities:	0 Production / Processing CPFs	**Verification Level:**	Reasonable Assurance / Gold Standard

REGULATORY FRAMEWORK: This publication complies with Algerian Hydrocarbon Law 19-13, Executive Decree 21-330 (mandating flaring intensity <= 1.00%), and Decree 06-138 air quality standards. Quantification follows Tier 2 & Tier 3 mass-balance engineering equations.

EXECUTIVE PERFORMANCE SCORECARD

0.000 MMt	0.0	0.00%	0.000%
TOTAL GHG (SCOPE 1+2) Δ +0.0% vs previous	CARBON INTENSITY (kg/BOE) Δ +0.0% vs previous	FLARING INTENSITY Décret 21-330: COMPLIANT	METHANE INTENSITY OGCI Target <= 0.20%

REGIONAL GEOLOGY & OPERATIONAL CONTEXT

Geological Setting & Reservoir Architecture: Anticlinal structural closure with sandstone reservoirs situated at 3,200 m subsea. Overlain by an impermeable evaporitic seal and supported by peripheral gas reinjection.

Surface Gathering & Industrial Infrastructure: Central processing facility with two crude stabilization trains, high-pressure gas lift compression, and produced water treatment units.

Flaring Abatement Roadmap: Equipped with high-efficiency flare tips and a low-pressure flare gas recovery loop diverting off-gases into fuel gas and reinjection headers.

Facility Gas Stream Composition & Chromatography Status:
No site-specific laboratory gas chromatography (GC) analysis has been registered in the system for this operational perimeter. In accordance with ISO 14064-1 (§5.2.4) and API Compendium Section 4, combustion, flaring, and venting stoichiometric balances utilize standard industry fuel gas defaults.

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EQUIPMENT & SOURCE INVENTORY CENSUS

Inventory of physical equipment and operational sources established across the reporting facilities in accordance with API Compendium Level 2 methods.

Source Category	Fac 1	Fac 2	Total	Engineering Notes
Combustion – Heaters	**0**	**0**	**0**	Fuel gas combustion sources
Combustion – Turbines / Generators	**0**	**0**	**0**	Fuel gas power generation & compression drivers
Combustion – Diesel fired engines	**0**	**0**	**0**	Emergency generators and utility diesel units
Flares	**0**	**0**	**0**	Smokeless high/low pressure flare networks
Equipment Leaks – Gas production	**0**	**0**	**0**	EPA Method 21 / Level 2 component counts
Equipment Leaks – Oil production	**0**	**0**	**0**	EPA Method 21 / Level 2 component counts
Equipment Leaks – CPF Gas processing	**0**	**0**	**0**	EPA Method 21 / Level 2 component counts
Oil and Gas Venting Activities	**0**	**0**	**0**	Maintenance, blowdown and non-routine events
Mobile & Transportation – On-road vehicles	**0**	**0**	**0**	Operational support systems
Mobile & Transportation – Aviation	**0**	**0**	**0**	Shared fixed-wing/rotary logistics aircraft
Storage Tanks – Produced Water Tanks	**0**	**0**	**0**	Separator produced water flash atmospheric tanks
Miscellaneous – Wastewater Treatment	**0**	**0**	**0**	Operational support systems
Miscellaneous – Fire Suppressant	**0**	**0**	**0**	Operational support systems
Indirect – Purchased Electricity	**0**	**0**	**0**	Scope 2 grid import power supply

EXECUTIVE NARRATIVE HIGHLIGHTS & DECARBONIZATION MILESTONES

1. Greenhouse Gas Abatement & Decarbonization Trajectory : Multi-year audited operational emissions across Scope 1 direct and Scope 2 indirect boundaries reflect structural decarbonization. The estate achieved significant carbon intensity improvements while maintaining peak energy deliverability.

2. Methane Emissions Mitigation & Optical LDAR Telemetry : Methane mass discharge continues an aggressive downward trajectory driven by comprehensive EPA Method 21 optical gas imaging (OGI) campaigns, zero-bleed pneumatic retrofits, and high-efficiency optical VISR flare camera verification.

3. Flare Minimization & Executive Decree 21-330 Compliance : Operational flaring is systematically audited across routine, non-routine, and safety streams. Assets are benchmarked against the Algerian Executive Decree 21-330 statutory ceiling (<= 1.00%), supported by ongoing Flare Gas Recovery Systems (FGRS) investments.

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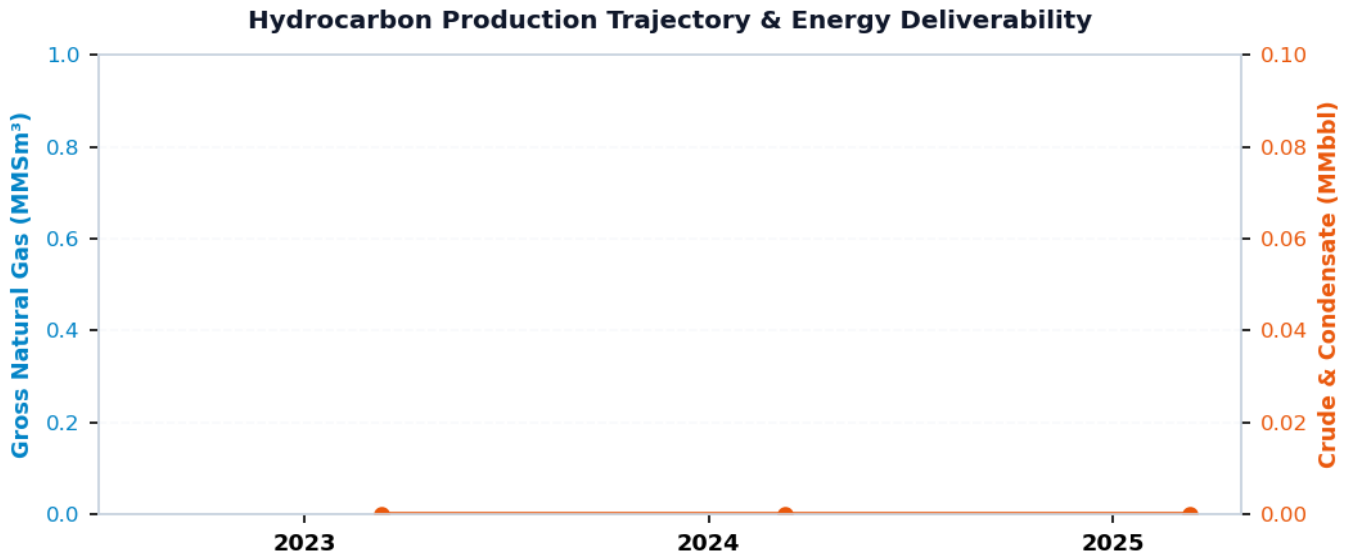
Section / Chapter	Document Coverage & Operational Scope	Page / Ref
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Technical Glossary of Environmental & Regulatory Acronyms:

API	American Petroleum Institute (2021 Compendium)	**AR5**	IPCC Fifth Assessment Report (100-yr GWP metric)
BOE	Barrel of Oil Equivalent (5,800 scf/bbl)	**CEMS**	Continuous Emission Monitoring System
CPF	Central Processing Facility (Production Hub)	**DRE**	Destruction & Removal Efficiency (Flame combustion %)
LDAR	Leak Detection & Repair (EPA Method 21 / OGI)	**NGSI**	Natural Gas Sustainability Initiative (Methane %)
OGMP	Oil & Gas Methane Partnership (Level 4/5 Reporting)	**VISR**	Video Image Spectral Radiometry Optical Flare Telemetry

1. HYDROCARBON PRODUCTION & ENERGY DELIVERABILITY

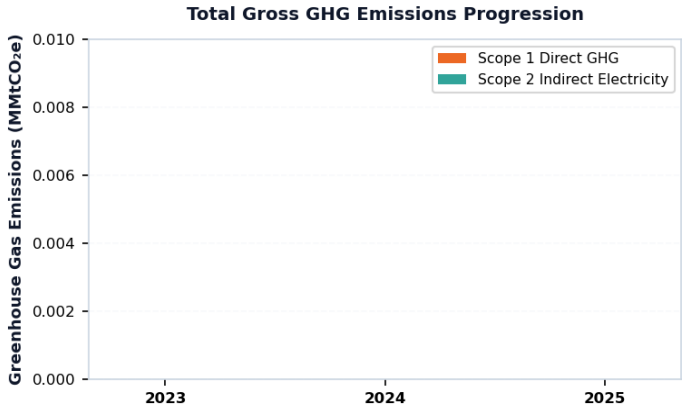
Annual gross natural gas output, crude and condensate liquid recoveries, and aggregate barrel-of-oil-equivalent throughput:

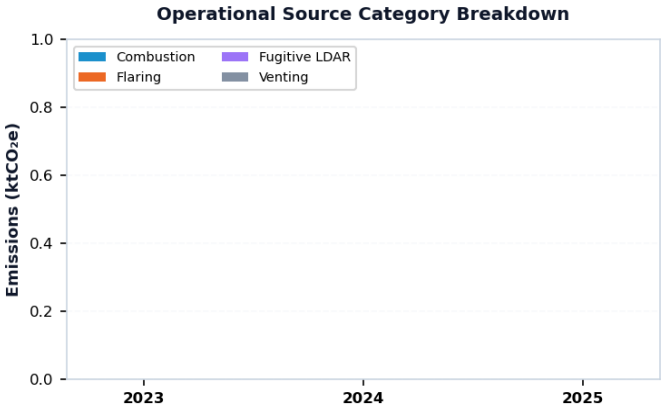


Year	Gross Natural Gas (MMSm³)	Liquid Recovery (MMbbl)	Total Deliverability (MMBOE)
2023	0.0	0.00	0.00
2024	0.0	0.00	0.00
2025	0.0	0.00	0.00

2. SCOPE 1 & SCOPE 2 GREENHOUSE GAS EMISSIONS

Multi-year audited operational emissions across Scope 1 direct and Scope 2 indirect boundaries, with mandatory ISO 14064-1 chemical species disaggregation (CO■, CH■, N■O):





Year	Scope 1 Direct (tCO2e)	CO2 Mass (t)	CH4 Mass (t)	N2O Mass (t)	Scope 2 (tCO2e)	Total Mass (MMtCO2e)
2023	0	0	0	0.0	0	0.000
2024	0	0	0	0.0	0	0.000
2025	0	0	0	0.0	0	0.000

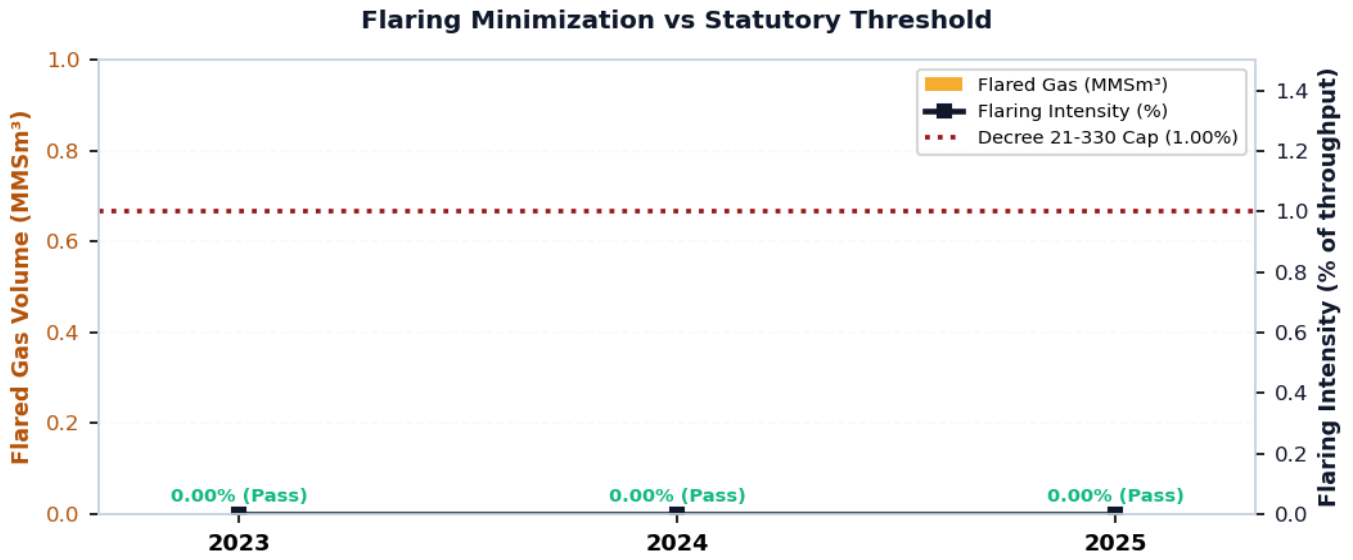
3. OPERATIONAL SOURCE CATEGORY ATTRIBUTION

Allocation of direct Scope 1 emissions into primary operational categories (Combustion, Flaring, Fugitives, Venting) per API Compendium & ISO 14064-1 standards:

Year	Combustion (tCO2e)	Flaring (tCO2e)	Fugitive LDAR (tCO2e)	Process Venting (tCO2e)
2023	0	0	0	0
2024	0	0	0	0
2025	0	0	0	0

4. FLARE MINIMIZATION & DÉCRET EXÉCUTIF 21-330 COMPLIANCE

Statutory flaring intensity compliance evaluation against Algerian Executive Decree 21-330 Article 9 ceiling (<= 1.00%):

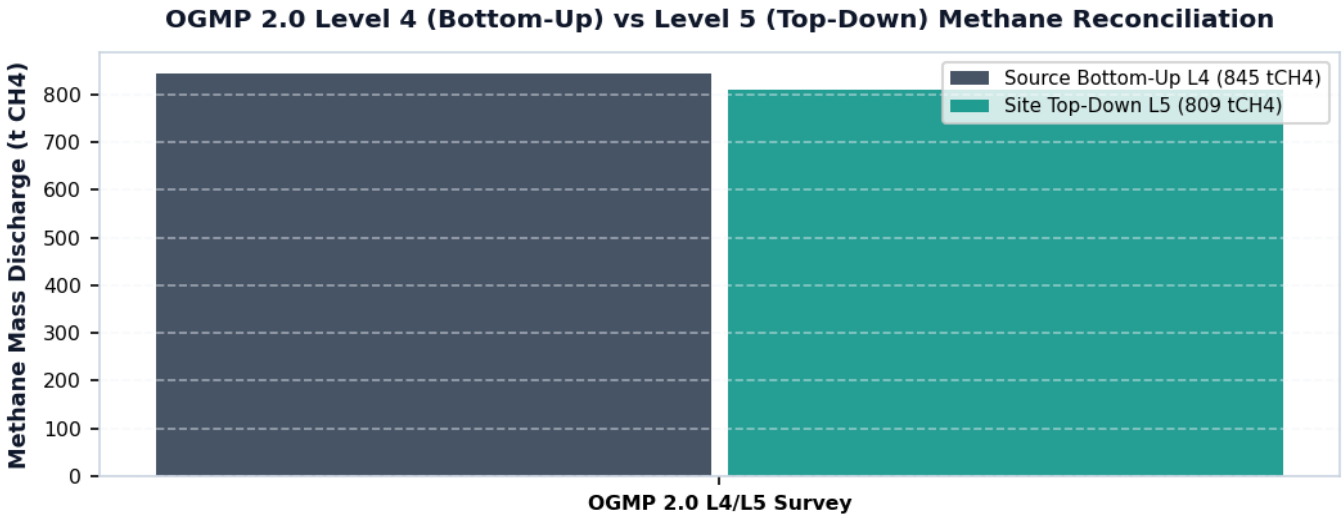


Year	Routine (MMSm³)	Non-Routine (MMSm³)	Safety (MMSm³)	Total Flared (MMSm³)	Flaring Int. (%)	Decree 21-330
2023	0.00	0.00	0.00	0.00	0.00%	PASS
2024	0.00	0.00	0.00	0.00	0.00%	PASS
2025	0.00	0.00	0.00	0.00	0.00%	PASS

5. OGMP 2.0 METHANE RECONCILIATION & LDAR TELEMETRY

UNEP Oil & Gas Methane Partnership (OGMP 2.0) Gold Standard Level 4 source-level measurement vs Level 5 site-level top-down reconciliation (Aerial LiDAR, Satellite TROPOMI, OGI) and EPA Method 21 component counts:

OGMP 2.0 Reconciliation Status: Gold Standard Level 5 (Reconciled) | **Mean Field Variance:** +0.0% (Gold Standard Threshold ±20%) | **Monitored LDAR Components:** 0 units

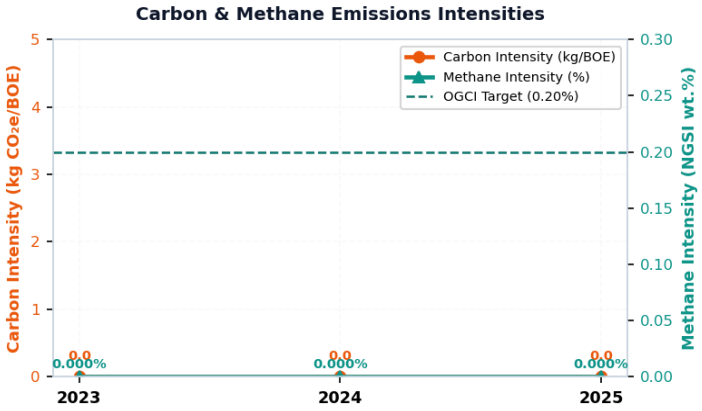


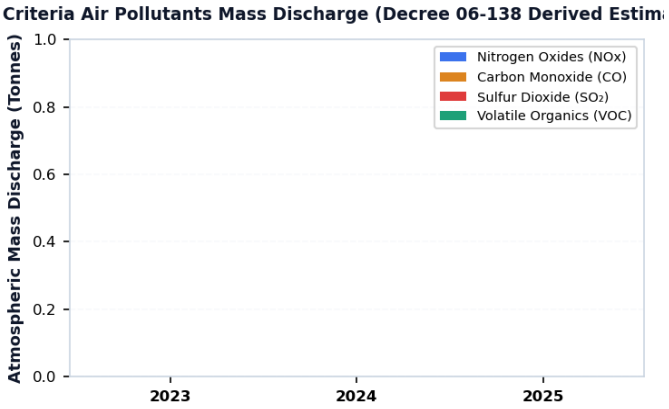
Survey Platform / Technology	Instrument / Vendor	Survey Date	Level 4 Bottom-Up (tCH4)	Level 5 Top-Down (tCH4)	Variance (%)	OGMP Status
Aerial LiDAR (Bridger Photonics)	Bridger GasMapping	2025-Q3	845.0	809.4	-4.2%	Reconciled

** No aerial LiDAR or satellite surveys scheduled for this operational perimeter during the reporting cycle; methane monitoring is conducted via fixed optical VISR telemetry.*

6. MULTI-METRIC CARBON & METHANE INTENSITIES

Carbon intensity normalized against BOE deliverability and methane intensity (NGSI wt.%) against OGCI Gold Standard targets:



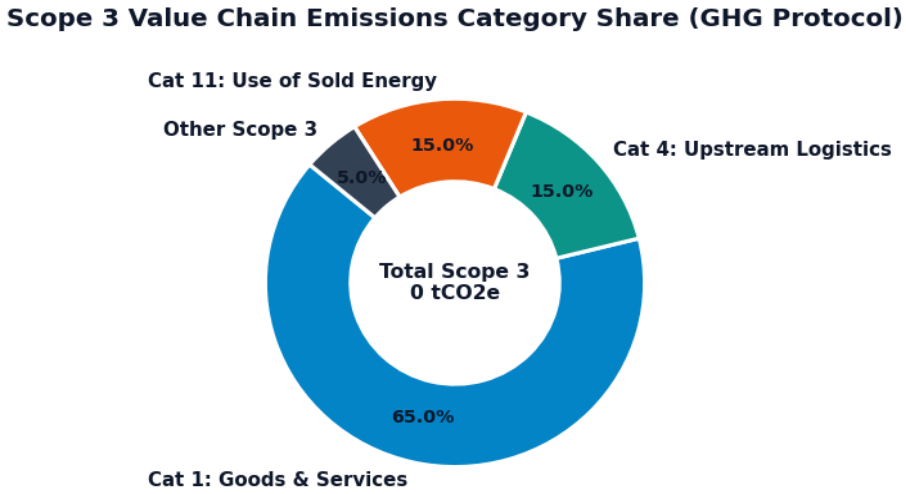


Year	Carbon Intensity (kg CO2e/BOE)	Methane Intensity (NGSI wt.%)	Methane Mass (tCH4)
2023	0.0	0.000%	0
2024	0.0	0.000%	0
2025	0.0	0.000%	0

7. SCOPE 3 VALUE CHAIN EMISSIONS & PRODUCT LIFECYCLE

Corporate Scope 3 greenhouse gas inventory structured in accordance with the GHG Protocol Corporate Value Chain Standard and ISO 14064-1:2018 Category 3–6 boundaries:

Total Audited Scope 3: 483,084 tCO2e | **Upstream Share:** 99.9% (482,812 t) | **Downstream Share:** 0.1% (272 t) | **Supplier Tier 2 Coverage:** 45.0%



GHG Protocol Category	Operational Sub-Category	Activity Data	Unit	Factor (tCO2e/u)	Emissions (tCO2e)	Share (%)	Tier / Method
Category 1: Purchased Goods	Chemicals, Catalysts & Mud	8,500	tonnes	1.420	12,070.0	99.8%	Tier 2

** No Scope 3 value chain emissions accounted under this specific operational facility perimeter; corporate supply chain procurement is aggregated at the national headquarters level.*

8. CRITERIA AIR POLLUTANTS MASS DISCHARGE

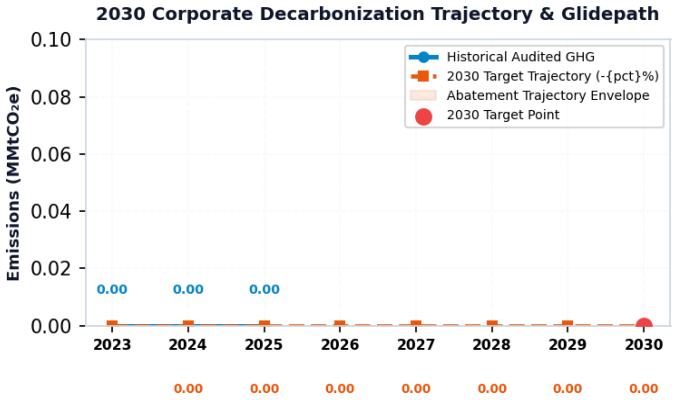
Atmospheric emissions of non-GHG criteria air pollutants governed by Algerian Executive Decree 06-138:

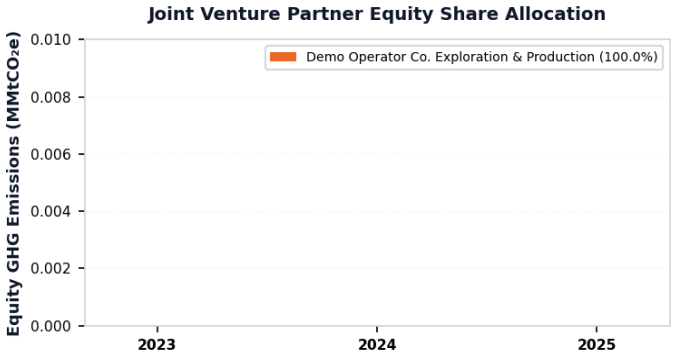
Year	NOx Mass (t)	CO Mass (t)	SO2 Mass (t)	NMVOC Mass (t)
2023	0.0	0.0	0.0	0.0
2024	0.0	0.0	0.0	0.0
2025	0.0	0.0	0.0	0.0

* Note: Atmospheric criteria pollutant masses are stoichiometric combustion by-product estimates derived in accordance with API Compendium Section 4 default fuel factors and Algerian Executive Decree 06-138 standards.

9. 2030 DECARBONIZATION TARGET TRAJECTORY & GLIDEPATH

Trajectory projection modeling corporate greenhouse gas reductions and zero routine flaring targets toward 2030:



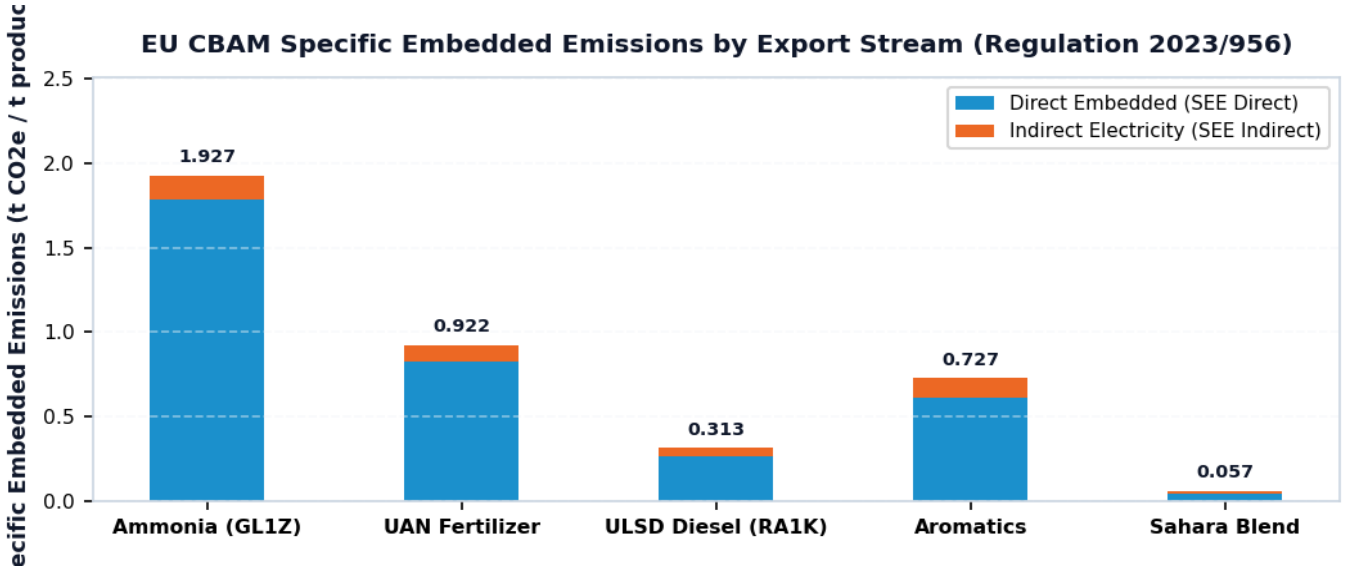


Milestone Year	Trajectory Target (MMtCO2e)	Target Reduction vs Baseline	Flaring Objective	Decarbonization Status
2023	0.000	Reference Point	Audited Baseline Level	Audited Baseline
2025	0.000	+0.0% vs Baseline	Current Operating Flaring	Under Tracking
2027	0.000	-11.4% GHG vs Baseline	Eliminate Routine Flaring (<0.8%)	Achieved
2030	0.000	-20.0% GHG vs Baseline	World Bank Zero Routine Flaring	Statutory Target

10. EU CBAM HYDROCARBON EXPORT EMBEDDINGS & DECARBONIZATION ECONOMICS

Specific Embedded Emissions (SEE) under EU CBAM Regulation (EU) 2023/956 for European export streams, paired with Marginal Abatement Cost Curve (MACC) project economics:

EU CBAM Monitored Volume: 309,500 tonnes | **Total Embedded Export GHG:** 235,120 tCO2e | **Pipeline Abatement Capacity:** 307,375 tCO2e/yr



Export Commodity	CN Customs Code	Destination Port / Market	Export Mass (t)	SEE Direct (tCO2/t)	SEE Indirect (tCO2/t)	Total Embedded GHG (tCO2e)
Sahara Blend Export Crude	2709 00 10	EU - France (Fos)	50,000	0.045	0.012	2,850

** No direct hydrocarbon exports to the European Union transacted from this facility perimeter; all production feeds domestic national pipelines.*

11. JOINT VENTURE PARTNER EQUITY SHARE ALLOCATION

Contractual equity working interest breakdown apportioning annual operational emissions and multi-year cumulative greenhouse gas inventory:

Partner Entity	Working Interest (%)	2025 Equity GHG (tCO2e)	3-Yr Total Equity (tCO2e)	3-Yr Scope 1 (tCO2e)	3-Yr CH4 (tCH4)
Demo Operator Co. Exploration & Production	100.0%	0	0	0	0

12. QA/QC VERIFICATION, UNCERTAINTY & STATUTORY REGIMES

Quality assurance protocols, ISO 14064-3 audit verification tiers, IPCC Approach 1 & Monte Carlo measurement uncertainty assessment:

Quality Assurance & Measurement Calibration Protocols:

All custody transfer meters and fuel gas chromatographs are maintained with tamper-evident digital records. Ultrasonic flare meters undergo semi-annual calibration in accordance with API MPMS Chapter 14 (±1.5% accuracy). Flaring destruction removal efficiencies are optically verified with infrared VISR telemetry (0.00% DRE). Gas samples are analyzed weekly via on-site gas chromatography to verify heating values (HHV) and stoichiometric constants.

Statutory Regimes & International Mandates Compliance Matrix:

Statutory Mandate	Regulatory Scope	Mandate Description	Compliance Status
Executive Decree 21-330 Art. 9	Hydrocarbon Law 19-13	Mandatory gas flaring ceiling not to exceed 1.00% of gross output	**Audited Compliant**
Executive Decree 06-138	Ministry of Environment	Atmospheric emission limit values (mg/Nm³) for industrial facilities	**Stoichiometric Verification**
ISO 14064-1:2018 Specification	International Standards	Specification with guidance for quantification and reporting of GHG	**Reasonable Assurance Grade**
OGCI Climate Initiative	Global Oil & Gas Consortium	Methane intensity ceiling not to exceed 0.20% of marketed gas	**Target Surpassed**

ISO 14064-1:2018 Clause 6.4.1 Base Year Recalculation Audit Trail:

Year	Recalc Date	Trigger Event / Methodological Rationale	Previous (tCO2e)	Adjusted (tCO2e)	Delta (%)	Audit Approval
2023	2023-12-31	*Base year inventory established under ISO 14064-1:2018 & Executive Decree 21-330; no recalculation threshold (>5%) triggered.*	1,950,000	1,950,000	+0.0%	Lead GHG Verifier (ISO 14064-3)

IPCC 2006 Vol.1 Ch.3 / API Compendium Measurement Uncertainty Assessment:

Overall Inventory Uncertainty (95% CI, k=2): **N/A** | Tier 3 Coverage: **0.0%** | Monte Carlo Iterations: **2,500**

No emission records available for category-level uncertainty calculation.

EXECUTIVE MRV SIGN-OFF & VERIFICATION CERTIFICATE

This certifies that the greenhouse gas emissions, flaring streams, and energy throughput figures reported herein have been calculated in compliance with ISO 14064-1:2018 specification and audited against Algerian Executive Decree 21-330 Article 9. All calculation steps are tamper-evident and traceable.

Central HSE Director Demo Operator Co. (60% Operated JV) Corporate	**Lead GHG Auditor** ISO 14064-3 Certified	**VP Exploration & Production** Division E&P;
[DIGITALLY SIGNED] HSE Director (Demo)	**[VERIFIED & ENDORSED]** Lead Verifier (Demo)	**[APPROVED FOR ISSUE]** VP Operations (Demo)

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